CINEMA TICKET RESERVATION AND REVENUE MANAGEMENT SYSTEM

Assembly Language Course Project

Course	Microcomputer Systems and Assembly Language 1301224
Academic Year	2026/2027
Semester	First
Project Type	Group Project - 3 students per group
Total Marks	30 marks across three phases
Assembler	MASM with Irvine32 Library
Instructors	Dr. Mohammed Khader

Project Theme: reservation processing, seat allocation, cancellation, pricing, searching, arrays, statistics, and formatted reporting in x86 Assembly.

SECTION 1: PROJECT OVERVIEW

1.1 Project Title

Cinema Ticket Reservation and Revenue Management System (CTRMS) - x86 Assembly Language Implementation

1.2 Project Description

You are required to design and implement a complete cinema ticket reservation system using x86 Assembly Language (MASM) with the Irvine32 library. The program will manage customer bookings, allocate seats, prevent duplicate seat reservations, calculate ticket prices according to movie and ticket category, support cancellation, display individual tickets and seating maps, and generate booking and revenue statistics.

The project is intentionally designed to require practical use of arrays, string processing, arithmetic operations, searching, validation, procedures, conditionals, and different loop structures while remaining suitable for implementation in an introductory microcomputer and assembly language course.

1.3 Learning Objectives

CLO Code	Description
S1	Develop assembly language programs for microcomputers to perform basic input/output operations.
S2	Implement and debug assembly language programs that perform arithmetic, string operations, searching, and memory manipulation tasks.
C1	Distinguish between different processor registers and use them appropriately in assembly language programming.

1.4 Project Value

Total Marks: 30 (divided across 3 phases)

SECTION 2: FUNCTIONAL REQUIREMENTS

2.1 Input Operations

The system manages three movie halls. Each movie has 20 seats, numbered from 1 to 20. A maximum of 60 active bookings can exist at one time.

Input	Format	Validation Rules
Customer Name	String	Maximum 20 characters; cannot be empty.
Booking ID	Integer (6 digits)	Must be between 100000 and 999999 and must be unique.
Movie Code	Integer (1-3)	1 = Movie A, 2 = Movie B, 3 = Movie C.
Seat Number	Integer (1-20)	Must be valid and not already reserved for the selected movie.
Ticket Category	Integer (1-3)	1 = Adult, 2 = Student, 3 = Child.

2.2 Processing Operations

Ticket Pricing Rules

Movie Code	Movie	Base Ticket Price (JOD)
1	Movie A	8
2	Movie B	7
3	Movie C	6

Ticket Category	Discount Rule
Adult	No discount
Student	1 JOD discount
Child	2 JOD discount

Final ticket price = Base ticket price - Category discount. All calculations use integer arithmetic; floating-point instructions are not required.

Calculations Required

Calculation	Formula / Rule
Final Ticket Price	Base movie price - ticket category discount
Movie Revenue	Sum of final prices for active bookings of that movie
Overall Revenue	Sum of final prices for all active bookings
Average Ticket Price	Overall revenue ÷ number of active bookings
Movie Occupancy	(Reserved seats × 100) ÷ 20; display as an integer percentage

Statistical Analysis Required

Statistic	Description
Total Active Bookings	Number of currently active reservations.
Movie Booking Count	Number of reserved seats for each movie.
Movie Revenue	Revenue generated by each movie.
Most Booked Movie	Movie with the highest number of active bookings.
Least Booked Movie	Movie with the lowest number of active bookings.
Average Ticket Price	Overall revenue divided by number of active bookings.
Ticket Category Distribution	Counts of Adult, Student, and Child tickets.
Occupancy Percentage	Integer occupancy percentage for each movie.

2.3 Output Operations

Main Menu

```
==================================================
 CINEMA TICKET RESERVATION & REVENUE SYSTEM
==================================================
1. Book New Ticket
2. Cancel Booking
3. Display Ticket
4. Display Seating Map
5. Booking & Revenue Statistics
6. Exit Program

Enter your choice (1-6): _
```

Ticket Format

```
==================================================
                    CINEMA TICKET
==================================================
Customer Name: Sara Ahmad
Booking ID: 345678
Movie: Movie A
Seat Number: 12
Ticket Category: Student
Base Price: 8 JOD
Discount: 1 JOD
--------------------------------------------------
Final Ticket Price: 7 JOD
==================================================
```

Seating Map Format

```
==================================================
                MOVIE A - SEATING MAP
==================================================
Legend: [O] Available   [X] Reserved

01[O] 02[O] 03[X] 04[O] 05[O]
06[O] 07[X] 08[O] 09[O] 10[O]
11[O] 12[X] 13[O] 14[O] 15[O]
16[O] 17[O] 18[O] 19[O] 20[O]

Reserved: 3 / 20
Occupancy: 15%
```

Statistics Format

```
==================================================
             BOOKING & REVENUE STATISTICS
==================================================
Movie A: 8 bookings   Revenue: 54 JOD   Occupancy: 40%
Movie B: 5 bookings   Revenue: 31 JOD   Occupancy: 25%
Movie C: 3 bookings   Revenue: 16 JOD   Occupancy: 15%
--------------------------------------------------
Total Active Bookings: 16
Overall Revenue: 101 JOD
Average Ticket Price: 6 JOD
Most Booked Movie: Movie A
Least Booked Movie: Movie C

TICKET CATEGORY DISTRIBUTION:
Adult   : ********  (8)
Student : *****     (5)
Child   : ***       (3)
==================================================
```

2.4 Technical Requirements

Control Structures (Must demonstrate all)

- At least 2 different loop structures (for example, LOOP instruction and a conditional/JMP-based WHILE-style loop).
- IF statements.
- IF-ELSE statements.
- Nested conditionals.

Procedures (Minimum 8 required; 12+ recommended)

Procedure	Purpose
Initialize	Clear arrays, booking records, seat maps, and counters.
DisplayMenu	Display the main menu and obtain the user choice.
AddBooking	Collect and validate all booking information.
ValidateBookingID	Validate six-digit format and uniqueness.
ValidateMovieCode	Validate movie code range 1-3.
ValidateSeat	Validate seat range and check availability.
CalculateTicketPrice	Determine base price, discount, and final ticket price.
FindBookingByID	Search the booking array for a matching active ID.
CancelBooking	Deactivate a booking and release its seat.
DisplayTicket	Display one formatted ticket by booking ID.
DisplaySeatMap	Show available/reserved seats for a selected movie.
CalculateStatistics	Scan active records and calculate counts, revenue, and occupancy.
DisplayStatistics	Display booking and revenue summary.
DisplayHistogram	Display ticket-category distribution using asterisks.

Data Types (Must use all)

- Strings (customer names).
- Integers / DWORD values (booking IDs, prices, revenue, counters).
- Bytes (movie codes, seat numbers, ticket categories, active flags).
- Characters (for seating-map symbols such as O and X).
- Arrays (booking records, seat maps, statistics counters).

Memory Management

- Use arrays or fixed-size records to store up to 60 bookings.
- Use a seat map for each movie to track available and reserved seats.
- Implement proper input validation before writing data to arrays.
- Use a fixed-size record format so that records can be accessed by index and offset.

Input / Output

- User-friendly prompts and clear instructions.
- Formatted ticket and statistical reports.

- Specific error messages for invalid input, duplicate IDs, full capacity, and reserved seats.
- Clear seating-map symbols and proper alignment.

SECTION 3: PROJECT PHASES

PHASE 1: PROPOSAL & ANALYSIS (7.5 marks)

Due Date: [November 5, 2026]

Deliverables

#	Deliverable	Description
1	Project Proposal Document	Problem statement, scope, objectives, and success criteria.
2	Problem Analysis Document	Input analysis, output analysis, process analysis, constraints, and assumptions.
3	Requirements Specification	Complete functional and non-functional requirements with CLO mapping.

Submission Format

- Single DOCX file named: Phase1_GroupX.docx (where X is your group number).
- Include all group members' names and IDs on the cover page.

Phase 1 Rubric (7.5 marks total)

Criteria	Excellent (90-100%)	Good (70-89%)	Satisfactory (50-69%)	Poor (0-49%)	Marks
Project Proposal (2.5)	Clear problem statement, scope, objectives, and measurable success criteria.	Good proposal with most elements.	Basic proposal with some missing elements.	Vague or missing proposal.	/2.5
Problem Analysis (2.5)	Thorough input/output/process analysis with all major constraints and assumptions.	Good analysis with minor omissions.	Basic analysis with gaps.	Minimal or incorrect analysis.	/2.5
Requirements Specification (2.5)	Complete functional and non-functional requirements with clear priorities and CLO mapping.	Good requirements with minor omissions.	Basic requirements with several gaps.	Incomplete or unclear requirements.	/2.5

What Students Must Submit for Phase 1

1. Project Proposal (2.5 marks)

- Project Title (0.25).
- Problem Statement (0.5) - explain what problem the cinema system solves.
- Scope (0.5) - clearly identify included and excluded features.
- Objectives (0.75) - minimum 3 specific and measurable objectives.
- Success Criteria (0.5) - explain how successful completion will be evaluated.

2. Problem Analysis (2.5 marks)

- Input Analysis Table (0.5) - all inputs with data types, ranges, and validation rules.
- Output Analysis Table (0.5) - all required outputs, formats, and purposes.
- Process Flow Diagram (0.75) - visual representation of booking, cancellation, display, and statistics flow.
- Constraints Identification (0.5) - memory, record capacity, integer arithmetic, and platform constraints.
- Assumptions (0.25) - assumptions made during analysis.

3. Requirements Specification (2.5 marks)

- Functional Requirements Table (1.0) - minimum 10 functional requirements with IDs and priorities.
- Non-Functional Requirements Table (1.0) - minimum 4 non-functional requirements.
- CLO Mapping (0.5) - show how the requirements support S1, S2, and C1.

PHASE 2: DESIGN (7.5 marks)

Due Date: [November 26, 2026]

Deliverables

#	Deliverable	Description
1	Algorithm Design	Flowcharts and pseudocode for all major procedures.
2	Data Structure Design	Booking record layout, seat arrays, statistics arrays, and variable allocation.
3	Procedure Specifications	Complete interface definitions for all major procedures.

Submission Format

- Single DOCX file named: Phase2_GroupX.docx (where X is your group number).
- Include all group members' names and IDs on the cover page.

Phase 2 Rubric (7.5 marks total)

Criteria	Excellent (90-100%)	Good (70-89%)	Satisfactory (50-69%)	Poor (0-49%)	Marks
Algorithm Design (2.5)	Clear flowcharts and pseudocode for all modules; logic handles normal and error paths.	Good algorithms with minor issues.	Basic algorithms with gaps.	Incomplete or incorrect algorithms.	/2.5
Data Structure Design (2.5)	Efficient fixed-size records, seat maps, and statistics arrays with clear offsets and sizes.	Good design with minor issues.	Basic structures present.	Poor or missing structure design.	/2.5
Procedure Specifications (2.5)	All 8+ procedures defined with inputs, outputs, register usage, and purpose.	Most procedures fully defined.	Some procedures defined.	Missing or unclear procedure specifications.	/2.5

What Students Must Submit for Phase 2

1. Algorithm Design (2.5 marks)

- Main Program Flowchart (0.75) - complete menu and control flow.
- Book Ticket Procedure Flowchart (0.75) - booking input, validation, seat check, price calculation, and storage.
- Validation / Search Flowchart (0.5) - ID uniqueness, seat availability, and input validation logic.
- Pseudocode for all procedures (0.5) - minimum 8 procedures.

2. Data Structure Design (2.5 marks)

- Memory Map Diagram (0.75) - show data segment, major arrays, and sizes.
- Booking Record Structure Definition (0.75) - define offsets and field sizes.
- Array Design (0.5) - show booking array, seat maps, and statistics arrays.
- Register Usage Convention Table (0.5) - specify intended use of EAX, EBX, ECX, EDX, ESI, and EDI.

3. Procedure Specifications (2.5 marks)

- Procedure Interface Table (1.0) - procedure name, inputs, outputs, modified registers, and purpose.

- Procedure Call Hierarchy Diagram (0.75) - show which procedures call which.
- Parameter Passing Convention (0.75) - explain whether registers, memory, or stack parameters are used.

PHASE 3: IMPLEMENTATION, TESTING & PRESENTATION (15 marks)

Due Date: [January 7, 2026]

Deliverables

#	Deliverable	Description
1	Complete ASM Code	Single .ASM file that compiles and runs without errors.
2	Test Cases & Results	Documented test cases showing expected versus actual results.
3	Video Presentation	Maximum 6 minutes; all group members must speak.

Submission Format

- Code: Project_GroupX.asm
- Testing Document: Phase3_Testing_GroupX.docx
- Video: Submit a shareable video link (for example, unlisted YouTube or Google Drive). Do not submit the video file unless instructed.

Phase 3 Rubric (15 marks total)

Criteria	Excellent (90-100%)	Good (70-89%)	Satisfactory (50-69%)	Poor (0-49%)	Marks
Code Implementation (6)	All required features work; clean structure; arrays, loops, validation, searching, pricing, cancellation, statistics, and formatting are correct.	Most requirements work with minor issues.	Basic functionality works.	Code does not compile or has major errors.	/6
Testing (4)	8+ well-documented tests; normal, boundary, duplicate-seat, cancellation, capacity, and error cases covered.	Good test coverage (5-7 tests).	Basic tests (3-4 tests).	Minimal or no testing.	/4
Presentation (5)	Clear explanation; demonstrates all features, key code, validation, arrays, register use, and CLO achievement; all members participate.	Good presentation with most features.	Basic presentation.	Poor or missing presentation.	/5

What Students Must Submit for Phase 3

1. Code Implementation (6 marks)

Requirement	Points
Code compiles without errors	1.0
Code runs without runtime crashes	0.5

Requirement	Points
All input validation works correctly	0.5
Unique booking ID and seat-availability checks work	0.5
Ticket price calculation is accurate	0.5
Cancellation releases the correct seat	0.5
Seating map displays correctly	0.5
Revenue and booking statistics are accurate	0.5
At least 2 different loop structures are used	0.5
At least 8 procedures are implemented	0.5
Arrays / fixed-size records are used correctly	0.5
Code comments and register usage are appropriate	0.5
Total	6.0

2. Testing (4 marks)

Test Case	Description	Points
TC1	Normal booking with valid data and price calculation.	0.5
TC2	Multiple bookings across all three movies; verify counts and revenue.	0.5
TC3	Boundary seats: seat 1 and seat 20.	0.5
TC4	Invalid input: booking ID, movie code, seat number, and category.	0.5
TC5	Duplicate booking ID and duplicate seat reservation.	0.5
TC6	Cancellation: cancel an active booking and verify seat becomes available.	0.5
TC7	Display ticket / search: existing ID and non-existing ID.	0.5
TC8	Maximum capacity: fill all 60 seats and attempt one additional booking.	0.5
Total		4.0

Testing Document Format (for each test case)

```
## Test Case [Number]: [Name]
Description: [What is being tested]
Input Data:
  - Customer Name: [value]
  - Booking ID: [value]
  - Movie Code: [value]
  - Seat Number: [value]
  - Ticket Category: [value]
Expected Output: [What should happen]
Actual Output: [What actually happened]
Status: [PASS/FAIL]
Screenshot: [Attach if applicable]
```

3. Video Presentation (5 marks)

Section	Time	Content	Points
Introduction	1 min	Project purpose, main features, and team introduction.	0.5

Section	Time	Content	Points
Code Walkthrough	2 min	Key procedures, loops, arrays, validation, searching, seat map, pricing, and register usage.	1.5
Live Demonstration	2 min	Normal booking, duplicate-seat error, ticket display, cancellation, seating map, and statistics.	2.0
Conclusion	1 min	Summary, lessons learned, and CLO achievement (S1, S2, C1).	1.0

Additional Video Requirements

- Maximum 6 minutes; a penalty may be applied for exceeding the limit.
- All group members must speak with approximately equal participation.
- Clear audio and readable screen recording.
- Show both the ASM code and the running program.
- Submit a shareable link as instructed.

SECTION 4: TESTING REQUIREMENTS (Detailed)

4.1 Mandatory Test Cases

Test Case 1: Normal Ticket Booking

```
Input:
  Name: Sara Ahmad
  Booking ID: 345678
  Movie Code: 1
  Seat Number: 12
  Ticket Category: 2 (Student)

Expected:
  Base Price = 8 JOD
  Discount = 1 JOD
  Final Price = 7 JOD
  Seat 12 for Movie A becomes reserved
  Active Booking Count increases by 1
  Movie A revenue increases by 7 JOD
```

Test Case 2: Multiple Bookings and Statistics

```
Input 1: ID 111111, Movie 1, Seat 1, Adult -> 8 JOD
Input 2: ID 222222, Movie 1, Seat 2, Child -> 6 JOD
Input 3: ID 333333, Movie 2, Seat 5, Student -> 6 JOD
Input 4: ID 444444, Movie 3, Seat 10, Adult -> 6 JOD

Expected:
  Movie A: 2 bookings, Revenue = 14 JOD, Occupancy = 10%
  Movie B: 1 booking, Revenue = 6 JOD, Occupancy = 5%
  Movie C: 1 booking, Revenue = 6 JOD, Occupancy = 5%
  Overall Revenue = 26 JOD
  Average Ticket Price = 6 JOD using integer division
```

Test Case 3: Boundary Values

```
Test A: Movie Code = 1, Seat = 1, Category = 1 -> valid
Test B: Movie Code = 3, Seat = 20, Category = 3 -> valid
Test C: Booking ID = 100000 -> valid
Test D: Booking ID = 999999 -> valid
```

```
Expected:
  All boundary values are accepted when seats are available and IDs are unique.
```

Test Case 4: Invalid Inputs

```
Test A: Booking ID = 12345 -> error and reprompt
Test B: Booking ID = 1234567 -> error and reprompt
Test C: Movie Code = 0 or 4 -> error and reprompt
Test D: Seat = 0 or 21 -> error and reprompt
Test E: Ticket Category = 0 or 4 -> error and reprompt
Test F: Empty customer name -> error and reprompt
```

Test Case 5: Duplicate Booking ID and Seat Collision

```
Step 1: Add ID 555555, Movie 2, Seat 7.
Step 2: Try to add another booking with ID 555555.
Expected: "ERROR: Booking ID already exists!"

Step 3: Use a new ID but select Movie 2, Seat 7 again.
Expected: "ERROR: Seat already reserved!" and reprompt for another seat.
```

Test Case 6: Cancellation and Seat Release

```
Step 1: Add booking ID 666666, Movie 3, Seat 14.
Step 2: Cancel booking ID 666666.
Expected:
  Booking becomes inactive.
  Movie 3 seat 14 becomes available.
  The booking is excluded from active statistics and revenue.
Step 3: Book Movie 3 seat 14 again with a new ID.
Expected: Booking succeeds.
```

Test Case 7: Display Ticket / Search

```
Step 1: Add booking ID 777777.
Step 2: Select Display Ticket and search for ID 777777.
Expected: Complete formatted ticket is displayed.

Step 3: Search for ID 888888 when it does not exist.
Expected: "ERROR: Booking ID not found!
```

Test Case 8: Maximum Capacity

```
Reserve all 20 seats for Movie A, all 20 seats for Movie B, and all 20 seats for Movie C.
Expected:
  60 active bookings are accepted.
  Each movie shows 100% occupancy.
  Any attempt to create another active booking must display:
  "ERROR: Cinema capacity reached!
```

SECTION 5: SUBMISSION GUIDELINES

5.1 General Submission Rules

- Each group submits once; only one member submits on behalf of the group.
- All phases are due by 11:59 PM on the specified due dates.
- Late submission penalty: 20% for one day late, 50% for two days late, then zero marks.
- A submission is considered complete only when all required files for that phase are provided.

5.2 Naming Conventions

Phase	File Name
Phase 1	Phase1_GroupX.docx
Phase 2	Phase2_GroupX.docx
Phase 3 Code	Project_GroupX.asm
Phase 3 Testing	Phase3_Testing_GroupX.docx
Phase 3 Video	Presentation_GroupX_Link.txt or LMS link field

(Replace X with your group number, for example Group1 or Group2.)

5.3 Code Header Format

```
;===============================================================
; Project: Cinema Ticket Reservation and Revenue Management System
; Course: Microcomputer Systems and Assembly Language 1301224
; Instructor: Dr. Mohammed Khader / Dr. Khaled Alhashash
; Group Members:
;   [Name 1] - [ID 1]
;   [Name 2] - [ID 2]
;   [Name 3] - [ID 3]
; Phase: 3 - Implementation
; Due Date: [Date]
; Description: Ticket booking, seat allocation, cancellation,
;              pricing, seating maps, searching, revenue statistics,
;              and ticket-category histogram.
; CLOs: S1 (I/O operations), S2 (arithmetic/string/memory/search),
;       C1 (register usage)
;===============================================================
```

5.4 Academic Integrity

- All work must be original. Plagiarism or unauthorized copying may result in zero marks for all involved parties.
- Use of AI tools (such as ChatGPT or GitHub Copilot) must be disclosed and cited according to instructor requirements.
- Group members are expected to contribute equally; peer evaluation may be requested.
- Students must be able to explain any code they submit during the presentation or oral questioning.

SECTION 6: GRADING SUMMARY

Total Marks: 30

Phase	Component	Marks
Phase 1	Proposal & Analysis	7.5
Phase 2	Design	7.5
Phase 3	Implementation, Testing & Presentation	15
TOTAL		30

CLO Achievement Summary

CLO	Description	Assessed in

CLO	Description	Assessed in
S1	Basic I/O operations	Phase 3: prompts, menus, formatted tickets, seat maps, and reports.
S2	Arithmetic, string, searching, and memory operations	Phases 2 and 3: pricing, names, arrays, seat search, record search, and statistics.
C1	Register functions	Phases 2 and 3: register convention and correct use of EAX, EBX, ECX, EDX, ESI, and EDI.

SECTION 7: FREQUENTLY ASKED QUESTIONS

Q1: Can we use a different assembler?

A: No. Use MASM in Microsoft Visual Studio with the Irvine32 library as provided in the course setup.

Q2: How many students per group?

A: Groups of 3 students.

Q3: What if the code does not compile?

A: Partial marks may be awarded for correct logic and documentation, but functionality marks require working code.

Q4: Can we add extra features?

A: Yes. Reasonable extra features may earn bonus credit up to 5% of the phase marks, at the instructors' discretion. Examples: edit booking, multiple show times, VIP seats, or file save/load.

Q5: What is the maximum number of active bookings?

A: 60 active bookings: 20 seats for each of 3 movies.

Q6: Can a seat number be reused for another movie?

A: Yes. Seat 5 in Movie A and seat 5 in Movie B are different seats. A seat is unavailable only when the same movie and seat number are already reserved.

Q7: Do we need floating-point arithmetic?

A: No. Use integer arithmetic. Average price and occupancy percentage may be displayed using integer division.

Q8: What happens when a booking is cancelled?

A: Mark the booking inactive and release the corresponding seat so it can be booked again.

Q9: How should statistics handle cancelled bookings?

A: Cancelled/inactive records must not be counted in active bookings, occupancy, ticket distribution, or revenue.

Q10: How is the video submitted?

A: Submit a shareable link according to the LMS instructions. Do not submit the raw video file unless requested.

SECTION 8: CONTACT INFORMATION

Instructor	Email	Office Hours
Dr. Mohammed Khader	m_khader@asu.edu.jo	Posted on Edugate
Dr. Khaled Alhashash	k_alhashash@asu.edu.jo	Posted on Edugate

APPENDIX A: Sample Complete Program Execution

```
==================================================
 CINEMA TICKET RESERVATION & REVENUE SYSTEM
==================================================
1. Book New Ticket
2. Cancel Booking
3. Display Ticket
4. Display Seating Map
5. Booking & Revenue Statistics
6. Exit Program

Enter your choice (1-6): 1

Enter customer name: Sara Ahmad
Enter booking ID (6 digits): 345678
Enter movie code (1-3): 1
Enter seat number (1-20): 12
Enter ticket category (1=Adult, 2=Student, 3=Child): 2

Booking added successfully!
Final Ticket Price: 7 JOD

Press any key to continue...

==================================================
 CINEMA TICKET RESERVATION & REVENUE SYSTEM
==================================================
Enter your choice (1-6): 3
Enter booking ID to display: 345678

==================================================
                   CINEMA TICKET
==================================================
Customer Name: Sara Ahmad
Booking ID: 345678
Movie: Movie A
Seat Number: 12
Ticket Category: Student
Base Price: 8 JOD
Discount: 1 JOD
--------------------------------------------------
Final Ticket Price: 7 JOD
==================================================

Press any key to continue...

Enter your choice (1-6): 4
Enter movie code (1-3): 1

==================================================
                MOVIE A - SEATING MAP
==================================================
Legend: [O] Available   [X] Reserved

01[O] 02[O] 03[O] 04[O] 05[O]
06[O] 07[O] 08[O] 09[O] 10[O]
11[O] 12[X] 13[O] 14[O] 15[O]
16[O] 17[O] 18[O] 19[O] 20[O]

Reserved: 1 / 20
Occupancy: 5%

Press any key to continue...
```

APPENDIX B: Checklist for Students

Before Submitting Phase 1

- Project proposal contains all required sections.
- Problem analysis includes input, output, process, constraints, and assumptions.
- Requirements specification includes functional and non-functional requirements.
- All group members' names and IDs appear on the cover page.
- DOCX file is named correctly: Phase1_GroupX.docx.

Before Submitting Phase 2

- Flowcharts for all major procedures are included.
- Pseudocode is provided for at least 8 procedures.
- Memory layout and booking record structure are complete.
- Seat-map arrays and statistics arrays are documented.
- Procedure interface table and call hierarchy are complete.
- Register usage and parameter passing are documented.
- DOCX file is named correctly: Phase2_GroupX.docx.

Before Submitting Phase 3

- Code compiles without errors and runs without crashes.
- All required booking, cancellation, display, seating-map, and statistics features work.
- Minimum 8 procedures are implemented.
- At least 2 different loop structures are demonstrated.
- Arrays / fixed-size records are used for booking storage.
- Duplicate booking IDs and duplicate seat reservations are rejected.
- All required input validation is present.
- Testing document contains at least the 8 mandatory tests.
- Video is maximum 6 minutes and all group members participate.
- All files are named correctly.

APPENDIX C: Sample Data Structure Definition

```
; Booking Record Structure (36 bytes per booking)
; Offset  Size    Field           Description
; 0       20      customerName    Customer name, null-terminated
; 20      4       bookingID       Unique six-digit booking ID
; 24      1       movieCode       1-3
; 25      1       seatNumber      1-20
; 26      1       category        1=Adult, 2=Student, 3=Child
; 27      1       activeFlag      1=active, 0=cancelled/unused
; 28      4       basePrice       Movie base price
; 32      4       finalPrice      Price after category discount
; Total: 36 bytes

; Suggested arrays
; bookings        BYTE 60*36 DUP(?)
; movieASeats     BYTE 20 DUP(0)
; movieBSeats     BYTE 20 DUP(0)
; movieCSeats     BYTE 20 DUP(0)
; movieCounts     DWORD 3 DUP(0)
; movieRevenue    DWORD 3 DUP(0)
; categoryCounts  DWORD 3 DUP(0)
```

APPENDIX D: Suggested Register Usage Convention

Register	Suggested Use
EAX	Arithmetic results, integer input/output, return values.
EBX	Record offsets, temporary base/index values.
ECX	Loop counters.
EDX	String addresses for Irvine32 WriteString; arithmetic support when needed.
ESI	Source/index pointer while scanning booking records.
EDI	Destination/index pointer or seat-map index.

Good Luck